

PPOM-1 OMM Lab #28

MET of the Hip, Tibia, Fibula, Subtalar and Calcaneus Regions

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Review TABLE (from Bates 13e) - Range of Motion of the Hip Joint and Potential Hypertonic Musculature involved:

BOX 23-16

Range of Motion of the Hip Joint

Hip Movement	Primary Muscles Affecting Movement	Patient Instructions
Flexion	Iliopsoas and rectus femoris (especially when knee is in extension)	"Bring your knee to your chest and pull it against your abdomen."
Extension	Gluteus maximus, gluteus medius, adductor magnus, and hamstrings (especially when knee is in extension)	"Lie face down, then bend your knee and lift it up." OR "Lying flat, move your lower leg away from the midline and down over the side of the table."
Abduction	Gluteus medius and minimus, tensor fascia latae (TFL)	"Lying flat, move your lower leg away from the midline."
Adduction	Adductor brevis, adductor longus, adductor magnus, pectineus, and gracilis	"Lying flat, bend your knee and move your lower leg toward the midline."
External Rotation	Internal and external obturators, quadratus femoris, and superior and inferior gemelli	"Lying flat, bend your knee and turn your lower leg and foot across the midline."
Internal Rotation	Gluteus medius and minimus, TFL, and some assistance from the adductors	"Lying flat, bend your knee and turn your lower leg and foot away from the midline."

Bickley, Lynn S., Bates' Guide to Physical Examination and History Taking, Thirteenth Edition, Wolters Kluwer, 2021, Chapter 23: Musculoskeletal System, See: Regional Joint Examinations Section, search for "BOX 23-16" - LINK:

<https://login.nyit.idm.oclc.org/login?url=https://premiumbasicssciences.lwwhealthlibrary.com/content.aspx?sectionid=248761200&bookid=2964#248761394>

MET for Hip Flexion Dysfunction – Prone

(DiGiovanna)

Ex: Right hip flexion dysfunction

Prone, physician stands on the side opposite to the dysfunction at level of patient's legs facing the patient

Treatment:

1. The physician's cephalad hand is placed on the patient's sacrum with his index finger at the posterior superior iliac spine of the side to be treated.
2. The physician's other hand reaches across the patient's legs and is placed anterior to the thigh above the knee.
3. The leg is extended at the hip until motion occurs at the ilium as monitored by the physician's cephalad hand.
4. The patient is instructed to push his leg downward toward the table. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully and then the physician repositions the patient into the new barrier.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



Eileen L. DiGiovanna, Christopher J. Amen, Denise K. Burns, An Osteopathic Approach to Diagnosis and Treatment, 4e, Wolters Kluwer 2021: Chapter 97: Muscle Energy of the Lower Extremity LINK:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=248815820&bookid=2969>

MET for Hip Extension Dysfunction – Supine

(DiGiovanna)

Ex: Right hip extension dysfunction

Supine, physician stands on the side of the dysfunction at level of patient's legs facing the patient

Treatment:

1. The physician's cephalad hand is placed on the patient's anterior superior iliac spine of the side to be treated.
2. The physician's other hand holds the patient's leg above the ankle.
3. The leg is flexed at the hip until motion is limited. Occasionally, the patient's knee will bend. The barrier is just before this occurs. The physician can support the patient's leg on his arm or shoulder.
4. The patient is instructed to push his leg downward toward the table. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully, and then the physician repositions the patient into the new barrier.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET for Hip Adduction Dysfunction – Supine

(DiGiovanna)

Ex: Right hip adduction dysfunction

Supine, closer to the opposite side of the table, physician stands at the foot of the table closer to the side to be treated

Treatment:

1. The physician's hand is placed on the patient's knee or ankle on the nontreated side.
2. The physician's other hand holds the patient's leg to be treated above the ankle.
3. The leg to be treated is abducted until the limit of motion is limited palpated. The physician can stand in a position between the patient's legs as a means of resisting adduction of the side to be treated.
4. The patient is instructed to push his leg medially toward the other leg. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully and then the physician repositions the patient into the new barrier.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET for Hip Abduction Dysfunction – Supine

(DiGiovanna)

Ex: Right hip abduction dysfunction

Supine, closer to the side of the table on the side to be treated, physician stands at the foot of the table closer to the side opposite to the side to be treated

Treatment:

1. The physician's hand is placed on the patient's knee or ankle on the nontreated side.
2. The physician's other hand holds the patient's leg being treated above the ankle.
3. The leg to be treated is flexed enough to clear the other leg and then adducted until the limit of motion is palpated.
4. The patient is instructed to push his leg laterally. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully, and then the physician repositions the patient into the new barrier.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET for Hip External Rotation Dysfunction – Supine

(DiGiovanna)

Ex: Right hip external rotation dysfunction

Supine, physician stands on the side of the dysfunction at level of patient's legs facing the patient

Treatment:

1. The patient's hip and knee are flexed to 90 degrees each.
2. The physician's cephalad hand stabilizes the patient's bent knee.
3. The physician's other hand pulls the patient's foot and leg laterally creating internal rotation.
4. The patient is instructed to **push his foot medially**. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully and then the physician repositions the patient into the new barrier.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET for Hip Internal Rotation Dysfunction – Supine

(DiGiovanna)

Ex: Right hip internal rotation dysfunction

Supine, physician stands on the side of the dysfunction at level of patient's legs facing the patient

Treatment:

1. The patient's hip and knee are flexed to 90 degrees each.
2. The physician's cephalad hand stabilizes the patient's bent knee.
3. The physician's other hand pushes the patient's foot and leg medially creating external rotation.
4. The patient is instructed to **push his foot laterally**. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully and then the physician repositions the patient into the new barrier.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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Review TABLE (from Bates 13e) - Range of Motion of the Knee Joint and Potential Hypertonic Musculature involved:

BOX 23-17

Range of Motion of the Knee Joint

Knee Movement	Primary Muscles Affecting Movement	Patient Instructions
Flexion	Hamstring group: biceps femoris, semitendinosus, and semimembranosus	"Bend your knee."
Extension	Quadriceps: rectus femoris, vastus medialis, lateralis, and intermedius	"Straighten your leg."

Bickley, Lynn S., Bates' Guide to Physical Examination and History Taking, Thirteenth Edition, Wolters Kluwer, 2021, Chapter 23: Musculoskeletal System, See: Regional Joint Examinations Section, SEARCH FOR "BOX 23-17" - LINK:

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MET of Knee Flexion Somatic Dysfunction

(DiGiovanna)

Ex: Right Knee Flexion Somatic Dysfunction

Supine, physician stands on the side of the dysfunction at level of patient's legs facing the patient.

Treatment:

1. The patient's knee is extended to the extent possible.
2. The physician's cephalad hand stabilizes the patient's bent knee.
3. The physician's other hand holds the patient's leg above the ankle.
4. The patient is instructed to bend his knee. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully, and then the physician repositions the patient into the new barrier.
7. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET of Knee Extension Somatic Dysfunction

(DiGiovanna)

Ex: Right Knee Extension Somatic Dysfunction

Prone, physician stands on the side of the dysfunction at level of patient's legs facing the patient.

Treatment:

1. The patient's knee is flexed to the extent possible.
2. The physician's cephalad hand stabilizes the patient's bent knee.
3. The physician's other hand holds the patient's leg above the ankle.
4. The patient is instructed to straighten his knee. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully, and then the physician repositions the patient into the new barrier.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET of Tibia External Rotation Dysfunction - Seated

(Nicholas)

Ex: Left Tibia Externally Rotated with Anteromedial Glide

Seated, physician stands facing the patient.

Diagnosis: External rotation: Free; Internal rotation: Restricted

Anteromedial glide: Free; Posterolateral glide: Restricted

Palpation: Tenderness over the anteromedial joint space

Treatment:

1. The patient sits with the legs hanging off the table and the physician facing the patient.
2. The physician grasps the lateral aspect of the patient's foot and ankle with one hand and the other hand contacts the medial tibial plateau to monitor motion (anteromedial and posterolateral glide) (Fig. 10.319).
3. The physician dorsiflexes the ankle and internally rotates the distal tibia to the edge of the restrictive barrier (white arrow, Fig. 10.320).
4. The physician instructs the patient to "turn your foot outward" (black arrow, Fig. 10.321), which externally rotates the tibia, while the physician applies an unyielding counterforce (white arrow).
5. This isometric contraction is maintained for 3 to 5 seconds, and then the patient is told to "stop" and "relax."
6. Once the patient has completely relaxed, the physician internally rotates the distal tibia to the edge of the new restrictive barrier (white arrow, Fig. 10.322).
7. Steps 4 through 6 are repeated three to five times or until motion is maximally improved.
8. The physician reassesses the motion of the tibia and the other related diagnostic components of the dysfunction to determine the effectiveness of the technique.

Nicholas A, Nicholas E, Atlas of Osteopathic Techniques, 3rd edition, Wolters Kluwer 2015: Chapter 10: Muscle Energy Techniques (**go to Chapter 10: Muscle Energy Techniques --> see Lower Extremity, and Search for your topic**) - **LINK:**

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(Fig. 10.320)



(Fig. 10.321)



(Fig. 10.322)

MET of Tibia Internal Rotation Dysfunction - Seated

(Nicholas)

Ex: Left Tibia Internally Rotated with Posterolateral Glide

Seated, physician stands facing the patient.

Diagnosis: External rotation: Restricted; Internal rotation: Free

Anteromedial glide: Restricted; Posterolateral glide: Free

Palpation: Tenderness over the anteromedial joint space

Treatment:

1. The patient sits with the legs hanging off the table and the physician facing the patient.
2. The physician grasps the medial aspect of the patient's foot and ankle with one hand and the other hand contacts the medial tibial plateau to monitor motion (anteromedial and posterolateral glide) (Fig. 10.327).
3. The physician dorsiflexes the ankle and externally rotates the distal tibia to the edge of the restrictive barrier (white arrow, Fig. 10.328).
4. The physician instructs the patient to "turn your foot inward" (black arrow, Fig. 10.329), which internally rotates the tibia, while the physician applies an unyielding counterforce (white arrow).
5. This isometric contraction is maintained for 3 to 5 seconds, and then the patient is told to "stop" and "relax."
6. Once the patient has completely relaxed, the physician externally rotates the distal tibia to the edge of the new restrictive barrier (white arrow, Fig. 10.330).
7. Steps 4 through 6 are repeated three to five times or until motion is maximally improved.
8. The physician reassesses the motion of the tibia and the other related diagnostic components of the dysfunction to determine the effectiveness of the technique.

Nicholas A, Nicholas E, Atlas of Osteopathic Techniques, 3rd edition, Wolters Kluwer 2015: Chapter 10: Muscle Energy Techniques (**go to Chapter 10: Muscle Energy Techniques --> see Lower Extremity, and Search for your topic**) - LINK:

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(Fig. 10.328)



(Fig. 10.329)



(Fig. 10.330)

MET of Posterior Fibular Head Dysfunction

(Nicholas)

Ex: Left Posterior Fibular Head

Supine or Seated, physician stands or sits at the side of dysfunction.

Treatment:

1. The patient lies supine (or sits with the lower legs off the table), and the physician stands or sits at the side of dysfunction.
2. The physician places the hand closest to the knee in the popliteal fossa so that the metacarpal-phalangeal joint of the index finger approximates the posterior proximal fibula (head) (Fig. 10.307).
3. The physician's other hand externally rotates, dorsiflexes, and everts the patient's lower leg/foot/ankle (white arrow, Fig. 10.308) until the fibular head meets the edge of its anterior restrictive barrier.
4. The physician instructs the patient to internally rotate, plantar flex, and invert the lower leg/foot/ankle (black arrow, Fig. 10.309) as the physician applies an unyielding counterforce (white arrow).
5. This isometric contraction is maintained for 3 to 5 seconds, and then the patient is told to "stop" and "relax."
6. Once the patient has completely relaxed, the physician externally rotates, dorsiflexes, and everts the patient's lower leg (foot and ankle) (white arrow, Fig. 10.310) until the fibular head meets its new anterior restrictive barrier.
7. Steps 4 to 6 are repeated three to five times or until there is no further improvement in the restrictive barrier.
8. The physician reassesses the position and motion of the fibular head to determine the effectiveness of the technique.

Nicholas A, Nicholas E, Atlas of Osteopathic Techniques, 3rd edition, Wolters Kluwer 2015: Chapter 10: Muscle Energy Techniques (**go to Chapter 10: Muscle Energy Techniques --> see Lower Extremity, and Search for your topic**) - LINK:
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(Fig. 10.307)



(Fig. 10.308)



(Fig. 10.309)

MET of Anterior Fibular Head Dysfunction

(Nicholas)

Ex: Left Anterior Fibular Head

Supine or Seated, physician stands or sits at the side of dysfunction.

Treatment:

1. The patient lies supine (or sits with the lower legs off the table), and the physician stands or sits at the side of dysfunction.
2. The physician contacts the anterolateral aspect of the fibular head with the thumb of the hand closest to the patient's knee (Fig. 10.311).
3. The physician's other hand internally rotates, plantar flexes, and inverts the patient's lower leg/foot/ankle (white arrow, Fig. 10.312) until the fibular head meets the edge of its posterior restrictive barrier.
4. The physician instructs the patient to externally rotate, dorsiflex, and evert the lower leg/foot/ankle (black arrow, Fig. 10.313) as the physician applies an unyielding counterforce (white arrow).
5. This isometric contraction is maintained for 3 to 5 seconds, and then the patient is told to "stop" and "relax."
6. Once the patient has completely relaxed, the physician internally rotates, plantar flexes, and inverts the patient's lower leg/foot/ankle (white arrow, Fig. 10.314) until the fibular head meets its new posterior restrictive barrier.
7. Steps 4 to 6 are repeated three to five times or until there is no further improvement in the restrictive barrier.
8. The physician reassesses the position and motion of the fibular head to determine the effectiveness of the technique.

Nicholas A, Nicholas A, Nicholas E, Atlas of Osteopathic Techniques, 3rd edition, Wolters Kluwer 2015: Chapter 10: Muscle Energy Techniques (**go to Chapter 10: Muscle Energy Techniques --> see Lower Extremity, and Search for your topic**) - **LINK:**
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(Fig. 10.311)



(Fig. 10.312)



(Fig. 10.313)

Review TABLE (from Bates 13e) - Range of Motion of the Ankle Joint and Foot and Potential Hypertonic Musculature involved:

BOX 23-19		
Range of Motion of the Ankle Joint and Foot		
Ankle and Foot Movement	Primary Muscles Affecting Movement	Patient Instructions
Ankle Flexion (Plantar Flexion)	Gastrocnemius, soleus, plantaris, and tibialis posterior	"Point your foot toward the floor."
Ankle Extension (Dorsiflexion)	Tibialis anterior, extensor digitorum longus, and extensor hallucis longus	"Point your foot toward the ceiling."
Inversion	Tibialis posterior and anterior	"Turn the sole of your foot inward or toward your midline."
Eversion	Fibularis longus and brevis	"Turn the sole of your foot outward or away from your midline."

Bickley, Lynn S., Bates' Guide to Physical Examination and History Taking, Thirteenth Edition, Wolters Kluwer, 2021, Chapter 23: Musculoskeletal System, See: Regional Joint Examinations Section, search for "BOX 23-19" - LINK:

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MET for Tibiotalar Dorsiflexion Dysfunction

(DiGiovanna)

Ex: Right ankle dorsiflexion dysfunction

Supine, seated or prone, physician at level with foot to be treated

Treatment:

1. The physician grasps the patient's ankle with one hand at the level of the malleoli.
2. The physician's other hand is placed over the dorsum of the patient's foot (Fig. 97-3).
3. The patient's foot is brought into plantar flexion to the barrier.
4. The patient is instructed to push his foot into dorsiflexion. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully and then the physician repositions the patient into the new barriers.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET for Tibiotalar Plantar Flexion Dysfunction

(DiGiovanna)

Ex: Right ankle plantar flexion dysfunction

Supine, seated or prone, physician at level with foot to be treated

Treatment:

1. The physician grasps the patient's ankle with one hand at the level of the malleoli.
2. The physician's other hand is placed under the plantar surface of the patient's foot.
3. The patient's foot is brought into dorsiflexion to the barrier.
4. The patient is instructed to push his foot into plantar flexion. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully, and then the physician repositions the patient into the new barriers.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET for Talocalcaneal Eversion Dysfunction

(DiGiovanna)

Ex: Right calcaneal eversion dysfunction

Supine, seated, or prone, physician at level with foot to be treated

Treatment:

1. The physician grasps the patient's **heel** with one hand.
2. The physician's other hand grasps the patient's **forefoot**.
3. The patient's calcaneus is **deviated medially** to the barrier (Fig. 97-4). (*note: think of coronal plane*)
4. The patient is instructed to **push his heel laterally**. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully, and then the physician repositions the patient into the new barriers.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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MET for Talocalcaneal Inversion Dysfunction

(DiGiovanna)

Ex: Right calcaneal inversion dysfunction

Supine, seated, or prone, physician at level with foot to be treated

Treatment:

1. The physician grasps the patient's **heel** with one hand.
2. The physician's other hand grasps the patient's **forefoot**.
3. The patient's calcaneus is **deviated laterally** to the barrier. (*note: think of coronal plane*)
4. The patient is instructed to **push his heel medially**. The physician provides isometric resistance for 3 to 5 seconds.
5. The patient is instructed to relax fully, and then the physician repositions the patient into the new barriers.
6. The procedure is repeated until normal range of motion is accomplished or approximated.



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Stanley Schiowitz, D.O., F.A.A.O. Department of Osteopathic Manipulative Medicine

(AY 2025-26) Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>